Dx7

Dialysis Exchange Dashboard

Product Requirements Document + System Architecture Document

Field	Value
Version	1.3
Date	February 2026
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Platform	Progressive Web App (PWA)
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DOCTRINE: Dx7 is a lens, not a ledger. It shows what the data allows and nothing more.

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PART A: ARCHITECTURAL CONTEXT

A1. Design Philosophy

Dx7 is a standalone operational system with a deliberate transition architecture. It functions independently while maintaining strict compatibility with a future canonical platform.

Core Principles

- Dx7 is an operational cache, not a system of record
- All patient data originates from HL7 (LIS is source of truth)
- API contracts must be stable, platform-agnostic, and swappable
- Multi-tenancy is structural from day one
- Clinical interpretation is explicitly excluded — data only

The Constraint That Defines Us

DOCTRINE: No color-coded risk. No labels. No interpretation. The nephrologist judges. Dx7 presents.

This is not a limitation — it is the competitive moat. What we refuse to do defines what we are.

A2. Transition Architecture

Dx7 is designed to transition to a future canonical platform via config-only change and zero UI rework.

Current State (Dx7 v1)

Partner Lab LIS → Platform Ingestion Module → Dx7 Backend (Own DB) → Dx7 PWA

Dx7 operates standalone with its own database. The Ingestion Module is architecturally separable.

Future State

Partner Lab LIS → Canonical Platform Ingestion → Dx7 (API Consumer)

Dx7 becomes read-through. Local DB deprecated. Config-only change. Zero UI rework.

Design Decision	Benefit
tenant_id on all tables	No ID translation at swap
API mirrors canonical shape	Backend swap is config, not rebuild
MD Notes owned by upstream	Accessed via API only
No business logic in frontend	UI survives backend replacement
Ingestion Module separable	Lifts to canonical platform unchanged

PART B: PRODUCT REQUIREMENTS

B1. Product Definition

Dx7 is a Progressive Web App (PWA) for dialysis clinic staff to view pre-session lab results.

Attribute	Value
Product Type	PWA (browser-based, Add Shortcut to Home Screen browser-dependent, mobile-friendly)
Primary User	Charge Nurse
Secondary Users	Shift Nurse, Nephrologist (MD)
Data Source	HL7 from LABExpress LIS (HCLab)
Deployment	Cloud-hosted, multi-tenant
Branding	White label (PL) + Private label (Clinic)
Device Target	Tablet or shared workstation

What Dx7 IS

- A read-only lens on lab results
- A workflow tool for dialysis shifts
- A session management interface (patient → chair → results)
- A place for MD notes per session

What Dx7 is NOT

- NOT a system of record (HL7/LIS is source of truth)
- NOT an EMR or clinical decision support
- NOT a dialysis machine integration (v1)
- NOT a patient-facing portal

B2. User Roles

Role	Description	Key Functions
Dx7 SysAd	Platform administrator	Manage tenants, global config
PL Admin	Partner Lab administrator	Manage clinics, PL branding, view all clinics
Clinic Admin	Dialysis center administrator	Manage users, clinic branding
Charge Nurse	Shift leader	Select patients, assign chairs, view results, export
Shift Nurse	Floor nurse	View shift patients, view results, view MD notes (read-only)

MD	Nephrologist	View results, write session notes

Role Permissions Matrix

Function	Charge Nurse	Shift Nurse	MD
Select shift	✓	✓	✓
Select patients for shift	✓	✗	✗
Assign chairs	✓	✗	✗
View results	✓	✓	✓
View MD Notes	✓	✓	✓
Write MD Notes	✗	✗	✓
Export/Print	✓	✗	✗

v1 Admin Scope

Dx7 v1 implements minimal admin capability. Admin interfaces are scoped to operational necessity for the validation phase. Expansion is deferred to Phase 2+ pending adoption confirmation.

Role	v1 Admin Screens	v1 Admin Method
PL Admin	User CRUD, Center (Clinic) CRUD	Web UI
Clinic Admin	User CRUD (own clinic only)	Web UI
Dx7 SysAd	Tenant setup, global configuration	Seed scripts / direct DB (no UI in v1)

v1 Explicitly Excludes:

- Tenant management UI (SysAd operations remain script-based)
- System-level admin console or dashboard
- Clinic-level configuration UI beyond user management
- Role management UI (roles are fixed in code for v1)

Rationale: Dx7 v1 serves a single Partner Laboratory with known configuration. Tenant setup occurs once at deployment. Admin UI investment is justified only after adoption validates the product.

B3. Screen Flow & Workflow

Login → Select Shift (1,2,3,4) → Patient Roster → Session View

Workflow Steps

- Nurse logs in → selects clinic (if multi-clinic access)
- Nurse selects shift (1, 2, 3, or 4)
- Nurse sees patient roster with two selection methods: SEARCH (type name → tick to add) or TICK (scroll roster → tick checkbox)
- Selected patient = session created for that shift
- Nurse assigns chair (number or letter, duplicates flagged)
- Nurse views results and selects which tests to review for this session
- MD views results → writes note per session
- End of shift → next shift starts fresh

B4. Functional Specifications

B4.1 Patient Roster

⚠ Patient records are created automatically from HL7. No manual patient creation in Dx7. Corrections must be made in LIS.

Function	Description	Notes
View all patients	List all patients for this clinic	From HL7 data (source of truth)
Search patient	Type name to filter	Instant search
Tick to select	Add patient to current shift	Creates session
Bulk select	Select multiple from search results	Efficiency for large shifts
View patient details	Name, birthdate, contact	Read-only
Deactivate patient	Soft delete	Admin only, history preserved

B4.2 Session Management

A SESSION is created when a patient is selected for a shift.

Field	Description
session_id	UUID
patient_id	FK to patient
session_date	Date of session
shift_number	1, 2, 3, or 4
chair	Number or letter (nullable)
assigned_by	User who created session

assigned_at	Timestamp

B4.3 Chair Assignment

Chair assignments are PER SHIFT. Each shift starts fresh. Assignments persist in session record for audit.

Function	Description	Notes
Assign chair	Enter number or letter	Free text entry
Duplicate warning	Flag if chair already assigned	Warning only, not blocked
Change chair	Update assignment	Audit logged
Chair persistence	Stored per session	For audit trail

B4.4 Results Display

⚠ NO color coding. NO good/bad labels. NO clinical interpretation. All lab-provided data displayed as pass-through.

Data Integrity Doctrine

Every value displayed in Dx7 is canonical and traceable to its HL7 source. Manual transcription from paper, PDF, Viber, or any non-HL7 channel is eliminated. If it did not arrive via HL7, it does not exist in Dx7. The MD can trust that what they see is what the laboratory released.

View	What It Shows
Current	Most recent result for each test
Last	Previous result (before current)
Comparison	Current vs Last side-by-side
Days Since	"32 days ago" — flag stale results
Historical	All available results (expandable)
Longitudinal	6-month structured lab review — all results for selected tests in a single continuous view without per-month navigation. Outlier context visible from layout.
Reference Range	Lab-provided range displayed as pass-through
Abnormal Flag	H/L flag from lab displayed as pass-through

PRE/POST BUN Separation

BUNPRE (pre-dialysis BUN) and BUNPOST (post-dialysis BUN) are always displayed as distinct line items. They are never collapsed into a single BUN entry. The ingestion layer preserves the pre/post distinction at the canonical layer by mapping to separate analyte identifiers. This distinction is clinically critical for URR and Kt/V adequacy calculations.

Source Lab Traceability

Every displayed lab value is traceable to its HL7 source via source_message_id. When results originate from multiple laboratories, the source lab is indicated per result. This ensures traceability to the originating HL7 message for audit defense and data integrity verification.

Test Selection

- All results received via HL7 are stored
- Charge Nurse selects which tests to view per session
- Test selection is UI state only — not persisted (default: all checked)

Lab Notes (NTE)

- Lab notes from HL7 are stored
- Display of lab notes controlled by PL Admin tenant setting

B4.5 Stale/Missing Results

When patient was sent to a different lab (no results at PL):

- Show last available result (even if old)
- Display days since: "Last result: 45 days ago"
- Visual flag for stale data
- NO manual entry — HL7 remains source of truth

This incentivizes clinics to send future specimens to PL.

B4.6 MD Notes

Function	Description	Notes
Write note	Free text entry	MD only
View note	See existing notes	All roles can view
Edit note	Update within 24 hours	MD only, audit logged
Note attachment	Per session	Not per patient

Notes stored in Dx7 database for now. Migrates to canonical platform when live. Session = Consultation in canonical terms.

B5. Branding

Two-Layer Branding

Layer	Name	Configured By	Shows Where
White Label	PL Branding	PL Admin	Footer, accent color
Private Label	Clinic Branding	Clinic Admin	Header, logo, name

Visual Layout

[CLINIC LOGO] Clinic Name [User Menu]

MAIN CONTENT AREA

[PL LOGO] Powered by LABExpress

B6. Reporting & Export

Parameter	Options
Patient	Individual OR Batch (all in current shift)
Period	Date range (from - to)
Tests	Select specific tests or all
Format	PDF, CSV

Who Can Export

- Charge Nurse — own clinic patients
- Clinic Admin — own clinic patients
- PL Admin — all clinics under tenant
- Shift Nurse — NO
- MD — NO (asks nurse to export)

Audit Defense Export

Dx7 supports a structured export format designed for PhilHealth audit defense. This output consolidates the data elements required to demonstrate clinical oversight and dialysis adequacy management.

Audit Defense Export includes:

- Lab trends per patient (longitudinal result history for selected tests)
- MD review timestamps (when MD notes were created per session)
- Kt/V and URR traceability (pre-BUN, post-BUN, calculated adequacy metrics with source result dates)

- Session history (patient, date, shift, chair assignment, attending MD)

This export does not include clinical interpretation, risk assessment, or compliance conclusions. It provides the raw structured data required for the clinic or Partner Laboratory to prepare audit documentation.

B7. UX Enhancements

Navigation

- Next/Previous patient — navigate without returning to roster
- Quick search — search from any screen
- Session recovery — return to same state if browser closes

Results View

- Days since "32 days ago" instead of just date
- Collapsible panels — Pre-session vs Post-session
- Stale flag — visual indicator for old results
- Test selection — nurse checks which tests to view (UI state, default all)

Round Efficiency Targets

Dx7 must support nephrologist round efficiency at scale. The following targets define the performance envelope for clinical adoption:

Target	Specification
Round throughput	20 patients reviewed in under 90 minutes
Patient-to-patient navigation	Sub-1-second load time between patients
Big Three visibility	Potassium (K+), Phosphorus, and Hemoglobin are structurally prominent on session open — visible without scrolling or expanding panels
Cognitive load	Results, MD notes, and export accessible from same view — no screen-hopping between sections

These are design constraints, not clinical requirements. They ensure the tool does not become a bottleneck in the nephrologist’s workflow.

PART C: SYSTEM ARCHITECTURE

C1. Architecture Overview

System Diagram

```
LABExpress LIS (HCLab)
   │ HL7 ORU Messages
   ▼
PLATFORM INGESTION MODULE (Architecturally Separable)
HL7 Listener → Normalizer → Canonical Result Objects
   │
   ▼
Dx7 API (Auth • RBAC • Sessions • Notes)
   │
PostgreSQL (Canonical Cache)
   │
   ▼
Dx7 PWA (Browser, Tablet)
```

The Ingestion Module is architecturally separable for future platform integration. It can be lifted to the canonical platform unchanged.

Technology Stack

Layer	Technology	Notes
Database	PostgreSQL	Multi-tenant with tenant_id
Backend	Developer preference: PHP (Laravel), Node.js, or .NET	Stack-agnostic — backend is disposable at platform swap
API	REST + JSON	Match canonical patterns
Frontend	React or Vue.js	Angular excluded — see constraint below
Ingestion	Separable module	Emits canonical result objects

Frontend Framework Constraint

Angular is excluded as a frontend option. The Dx7 PWA is the permanent strategic asset that survives backend replacement at canonical platform transition. The frontend must maintain design system continuity with the canonical platform's component ecosystem. React and Vue.js share component patterns, styling approaches, and design system tooling that enable seamless

visual integration. Angular's opinionated framework architecture, different component model, and separate design system tooling create a structural incompatibility at the design system layer — exactly the layer that must be seamless across the platform family.

Dx7 is the first reference implementation in the x7 product family. The frontend framework choice propagates to all future verticals. This decision is architectural, not preferential.

C2. Database Schema

All tables include tenant_id for multi-tenancy. Schema mirrors canonical platform patterns.

TENANTS

id, name, code, logo_url, footer_text, primary_color, is_active, created_at, updated_at

CLIENTS (Dialysis Clinics)

id, tenant_id (FK), name, code, logo_url, address, is_active, created_at, updated_at

USERS

id, tenant_id (FK), client_id (FK, nullable for PL Admin), email, password_hash, name, role (enum: sysad|pl_admin|clinic_admin|charge_nurse|shift_nurse|md), is_active, created_at, updated_at

PATIENTS

id, tenant_id (FK), client_id (FK), lis_patient_id, name, birthdate, gender, contact_number, is_active, created_at, updated_at

UNIQUE INDEX (tenant_id, client_id, lis_patient_id)

SESSIONS

id, tenant_id (FK), client_id (FK), patient_id (FK), session_date, shift_number (1–4), chair, assigned_by (FK), assigned_at, created_at, updated_at

UNIQUE INDEX (patient_id, session_date, shift_number)

RESULTS (Operational Canonical Cache)

id, tenant_id (FK), patient_id (FK), accession_id, test_code, test_name, result_value, result_unit, reference_range, abnormal_flag, result_date, result_time, source_message_id, created_at, updated_at

INDEX (tenant_id, patient_id, test_code, result_date)

RESULTS table represents an operational canonical cache in v1 deployment topology. When canonical platform ingestion is live, Dx7 reads canonical result objects via API and this cache becomes optional or read-through.

MD_NOTES

id, tenant_id (FK), session_id (FK), md_user_id (FK), note_text, created_at, updated_at

CHAIR_AUDIT

id, session_id (FK), chair_old, chair_new, changed_by (FK), changed_at

C3. API Contracts

REST API with JSON. All endpoints require authentication except login.

Authentication

POST /api/auth/login — Request: { email, password } → Response: { token, user, tenant, client }

Patients

GET /api/patients?client_id=UUID&search=name&status=active|stale|nodata

GET /api/patients/:id

Sessions

GET /api/sessions?client_id=UUID&date=YYYY-MM-DD&shift=1

POST /api/sessions { patient_id, session_date, shift_number, chair }

POST /api/sessions/bulk { patient_ids: [...], session_date, shift_number }

PATCH /api/sessions/:id { chair }

Results

GET /api/results/current/:patient_id

GET /api/results/history/:patient_id/:test_code?from=DATE&to=DATE

MD Notes

GET /api/notes?session_id=UUID

POST /api/notes { session_id, note_text }

PATCH /api/notes/:id { note_text }

Export

POST /api/export { patient_ids, from_date, to_date, test_codes, format }

C4. Ingestion Module Requirements

The Ingestion Module is a separable component that receives HL7 messages and emits canonical result objects. It is not Dx7 logic — it is platform infrastructure deployed alongside Dx7 in v1.

Core Requirements

- Emits canonical result objects to Dx7 API / results cache
- Idempotent ingestion — safe to replay messages
- Restart-safe processing — no message loss on service restart
- Duplicate detection — handles re-transmitted messages gracefully
- Raw HL7 retention — original message stored for traceability
- Secure transport — no unencrypted HL7 over public internet
- PRE/POST BUN preserved as distinct canonical analytes — BUNPRE and BUNPOST mapped to separate identifiers at ingestion, never collapsed

Mapping Specification

Field-level HL7 mapping is defined in a separate HL7 Mapping Specification document, validated against real HCLab ORU samples. The Ingestion Module implements the approved mapping without invention.

Transport Options

Option	Topology	When To Use
A: VPN Tunnel	HCLab → VPN → MLLP Listener → Ingestion Module	LABExpress IT can establish tunnel
B: Edge Relay	HCLab → MLLP → Edge Relay → HTTPS → Ingestion Module	VPN not feasible

Edge Relay Constraints (If Option B)

- Receive MLLP from HCLab (local network only)
- Persist to local spool for retry/resilience
- Forward via HTTPS — no transformation
- No business logic — receive and forward only
- Not a system of record

C5. Security

- JWT tokens with 8-hour expiry (shift length)
- Role-based access control (RBAC)
- tenant_id enforced on ALL queries
- client_id enforced for clinic-level users
- HTTPS only (TLS 1.2+)
- Audit logging for all write operations
- Session recovery with secure token storage

PART D: SCREEN MOCKUPS

ASCII mockups included for reference. Final UI to match canonical platform design system.

D1. Login

[Dx7 Logo]

Dialysis Exchange Dashboard

[Email]

[Password]

[LOGIN]

D2. Shift Selection

[Clinic Logo] Clinic Name [User ▼]

Select Shift for Today

SHIFT 1 (6:00 AM, 12 pts) | SHIFT 2 (10:00 AM, 15 pts) | SHIFT 3 (2:00 PM, 10 pts) | SHIFT 4 (6:00 PM, 8 pts)

[PL Logo] Powered by LABExpress

D3. Patient Roster

☐	Chair	Patient Name	Last Result	Status
☑	A1	DELA CRUZ, Juan	1 day ago	● Ready
☑	A2	SANTOS, Maria	1 day ago	● Ready
☑	B1	REYES, Pedro	35 days ago	⚠ Stale
☐	—	LOPEZ, Jose	45 days ago	⚠ Stale
☐	—	BAUTISTA, Carlos	Never	○ No data

D4. Results View

Test	Current	Last	Flag
Potassium	5.4 mmol/L	5.2 mmol/L	
BUN PRE	19.82	18.50	H
BUN POST	6.12	5.90	
Creatinine	1072.67	980.00	H
Sodium	136.90	138.00	
Phosphorus	2.65	2.10	H

Hemoglobin	10.5 g/dL	10.2 g/dL	

Flag column shows H/L from lab (pass-through). Reference range available on drill-down. BUN PRE and BUN POST are always separate line items. Source lab indicated per result via source_message_id.

PART E: TRANSITION PLAN

E1. Dx7 Standalone → Canonical Platform Integration

Current State (Dx7 v1)

- Dx7 has its own database (operational canonical cache)
- Platform Ingestion Module feeds Dx7 directly
- MD Notes stored in Dx7

Future State (When Canonical Platform Live)

- Canonical platform ingestion receives HL7 from LIS
- Dx7 becomes API consumer of canonical platform
- Dx7 results cache becomes optional or read-through
- MD Notes migrate to canonical platform (session = consultation)

Transition Steps

- Canonical platform goes live with HL7 ingestion
- Configure Dx7 to consume canonical API (feature flag)
- Run parallel: verify data matches
- Migrate MD Notes from Dx7 to canonical platform
- Disable Dx7 database writes (or keep as read-through cache)

Design Decision	Benefit
Same tenant_id pattern	No ID translation needed
Same client_id pattern	Clinic mapping preserved
API mirrors canonical shape	Swap is config, not rebuild
Session = Consultation	MD Notes map cleanly
Ingestion Module separable	Lifts to platform unchanged

APPENDIX A: DIALYSIS PANELS & GLOSSARY

A1. Dialysis Test Panels (LABExpress Pricing)

Panel	Code	Tests Included	Price
Quarterly	RQL / Panel 05	Hemoglobin, Hematocrit, Albumin	₱490
Monthly Basic	RML 1 / Panel 10	CBC, Pre/Post BUN, URR/Kt/V, Cr, Albumin	₱700
Monthly Full	RML 2 / Panel 12	Panel 10 + P, UA, SGPT, Na, K, Ca	₱770
Comprehensive	Panel 15	Panel 12 + ALP, FBS, Lipid Profile	₱945

A2. Glossary

Term	Definition
Dx7	Dialysis Exchange Dashboard
PL	Partner Laboratory (e.g., LABExpress)
Client	Dialysis clinic served by PL
Tenant	PL instance in multi-tenant system
Session	Single dialysis session (patient + date + shift)
Shift	Time slot (1–4 per day)
Chair	Dialysis station assignment
HL7	Health Level 7 messaging standard
ORU	Observation Result message (HL7)
PWA	Progressive Web App
RML	Routine Monthly Labs
RQL	Routine Quarterly Labs

APPENDIX B: ARCHITECTURAL INVARIANTS

This appendix defines non-negotiable architectural boundaries for Dx7. These invariants are not guidelines — they are structural constraints that must be preserved regardless of implementation decisions, optimization pressures, or future feature additions.

Violation of any invariant requires explicit, documented approval and architectural review.

1. Trust Boundary

INVARIANT: The PWA is a presentation layer only and is treated as an untrusted client.

The browser runtime is not a secure environment. All user input, client state, and browser-side logic must be treated as potentially compromised.

Implications:

- No authentication tokens stored in accessible JavaScript memory
- No business rule evaluation in the client
- No tenant/branch scoping decisions made client-side
- "Add Shortcut to Home Screen" is browser-dependent, not native app trust

2. API Boundary Doctrine

INVARIANT: The API exists as the governance and enforcement boundary, not as an integration convenience.

The API is the single point of control for all system behavior. It exists even when the PWA is the only client.

The API is responsible for:

- Authentication (identity verification)
- Authorization (permission enforcement)
- Tenant and branch scoping (isolation guarantee)
- Audit logging (compliance trail)
- Business rule evaluation (consistency guarantee)
- Input validation (security boundary)

Direct database access from any client — including internal tools — is prohibited.

3. Tenant Isolation

INVARIANT: Tenant and branch isolation must be enforced in the service layer and fail-closed at the database layer. Client-side enforcement is prohibited.

Implementation requirements:

- Service layer injects tenant_id into all queries based on authenticated session
- Application code never constructs tenant_id from client input
- Database layer enforces row-level security or equivalent as backstop
- Missing tenant context must fail the request, not default to any tenant
- Cross-tenant queries require explicit, audited system-level authorization

4. Audit Path Guarantee

INVARIANT: All auditable actions must pass through the service/API layer to ensure complete, append-only audit trails.

Requirements:

- Every write operation (create, update, delete) generates an audit record
- Audit records are append-only — no updates or deletes permitted
- Audit records include: timestamp, user_id, tenant_id, action, entity, before/after state
- Audit logging cannot be bypassed by any code path, including batch operations
- Direct database modifications (migrations, fixes) must generate manual audit entries

5. Offline Reconciliation

INVARIANT: Offline actions are reconciled through idempotent service endpoints. The server is the sole authority for conflict resolution.

v1 assumes online-first environment. Offline queue implementation deferred unless field conditions require.

Requirements:

- All write endpoints must be idempotent (safe to retry)
- Client includes request_id or equivalent for deduplication
- Server determines final state in all conflict scenarios
- Client must accept server state after sync, even if different from local expectation
- Offline queue is a convenience, not a source of truth

6. Business Rule Residency

INVARIANT: Dx7 business rules are centralized and evaluated server-side to ensure consistency across all present and future clients.

This includes:

- Session creation rules (who can create, when, constraints)
- Chair assignment rules (duplicate detection, validation)
- MD Note permissions (who can write, edit windows, audit)
- Export permissions (role-based, tenant-scoped)
- Result display rules (what to show, stale thresholds)

Client may implement optimistic UI, but must defer to server response.

7. HL7 Ingestion Boundary

INVARIANT: HL7/MLLP is unencrypted by design. Direct transmission over public internet is prohibited. All normalization, validation, and persistence occur in the Ingestion Module.

Confirmed:

Item	Status
HCLab output protocol	MLLP (confirmed)
Transmission mode	Real-time push (not batch/file)

Cloud target	DigitalOcean

⚠ DISCOVERY REQUIRED: Can LABExpress network establish a VPN/WireGuard tunnel to DigitalOcean? Ask LABExpress IT.

Prohibited:

- Public MLLP listener (unencrypted PHI over internet)
- FTP-based integration
- Manual file upload as primary path
- Business logic in edge relay
- Edge relay as system of record

Summary: Layer Responsibilities

Layer	Trust Level	Responsibilities	Prohibitions
PWA (Client)	Untrusted	Presentation, input capture, optimistic UI	Auth decisions, rule evaluation, tenant scoping
API (Service)	Trusted	Auth, authz, tenant scope, audit, rules, validation	Direct DB exposure, client trust, audit bypass
Database	Backstop	Storage, row-level security, referential integrity	Direct client access, business logic
Ingestion Module	Trusted	Parse, normalize, validate, emit canonical objects	Business logic beyond mapping, being Dx7 logic
Edge Relay	Transport only	Receive MLLP, spool, forward HTTPS	Transformation, state, being system of record

The API is not a pass-through. It is the control plane.

The Ingestion Module is not Dx7 logic. It is platform infrastructure.

These invariants are structural, not stylistic. Deviation requires architectural review.

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